

Acceleration and variability in vocabulary growth for young children and language models

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Vocabulary growth

Theoretical: Vocabulary is a key nexus of language, social cognition, and conceptual development.

Practical: Variation in vocabulary growth is a key precursor of academic outcomes.

Methodological: Vocabulary count has a property nearly no other psychological metric has: a true zero, so we can explore the psychometrics of learning.

Comparative: We can explore how LLMs relate to children using dense data about child outcomes.

Wordbank Data

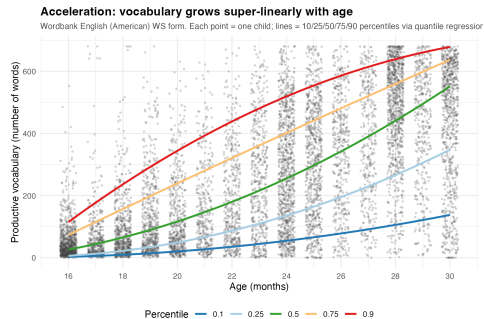
MacArthur–Bates Communicative Development Inventory (CDI): parent-report checklist of ~ 680 words.

We focus on production (less noisy).

Wordbank: open repository of CDI administrations. ~ 30 languages, $\sim 10^5$ children. We use both:

- **Cross-sectional:** one snapshot per child
- **Longitudinal:** ≥ 2 admins per child, lets us track per-child growth

Large N , parent-report calibrated to direct testing, cross-linguistically validated.



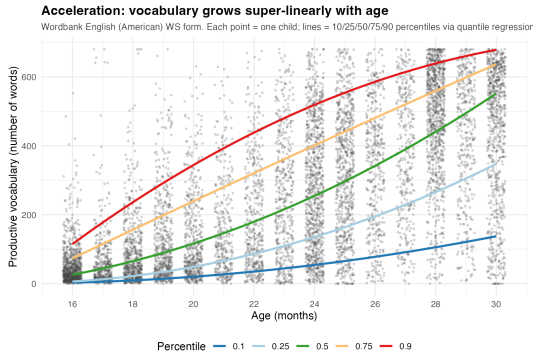
One child = one point. Vocabulary $\times$ age in English (American) WS.

Two signatures of vocabulary growth

Signature 1: acceleration. Each quantile curve is concave up: doubling age more than doubles vocabulary. The growth rate accelerates.

Signature 2: variability. The quantile lines fan out: between-child differences in vocabulary grow with age, not shrink.

These signatures are remarkably robust: replicate across languages, samples, and forms. Any model of early word learning must explain both.



Is growth pure accumulation?

Each child has input rate r_i ; word j has difficulty δ_j . A word is acquired when enough tokens accumulate.

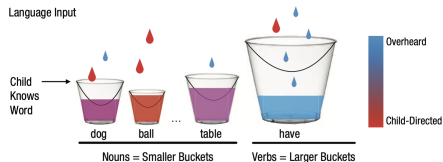
Derived from cross-situational word learning and other associative/statistical learning ideas.

In the literature:

- McMurray (2007) — pure rate $\times$ time
- Mitchell & McMurray (2009) — + leverage feedback
- Hidaka (2013) — cumulative-Weibull; rate change
- Mollica & Piantadosi (2017) — per-word Gamma
- Kachergis et al. (2021) — IRT Rasch + α_i

Acceleration: Some models have changes in efficiency to create acceleration, others do not.

Variation: Between-child variation only in baseline (r_i or α_i). No variation in efficiency.



Each child hears tokens at rate $r_i \cdot p_j$; word j is acquired when its bucket fills past δ_j .

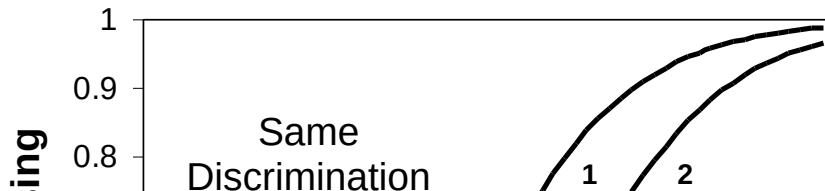
Item response theory (IRT)

IRT describes a family of models of binary data (like CDI data).

We use the Rasch model, a logistic model where each item has a single difficulty parameter δ , and each person has a single ability parameter θ .

Probability of success on an item is then just:

$$P(w_{i,j} = 1 | \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}$$



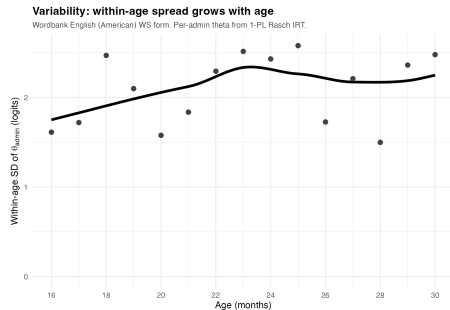
A puzzle: input matters less than supposed

The accumulator view predicts variation in growth comes from variation in input quantity. Coffey & Snedeker (2026) meta-analyzed 71 studies, 4,760 children:

R^2 of input on outcome = 0.04–0.07.

Caregiver speech quantity explains 4–7% of variance in language outcome — a real but small effect.

Hypothesis: Variability comes from *how children accelerate*, not how much input they hear.



Between-child $SD(\theta)$ grows from ~ 1.7 to ~ 2.3 logits between 16 and 30 mo. If input explains $\sim 5\%$, the rest comes from somewhere.

This work: modeling individual variability in acceleration

We build a Bayesian model to estimate three things directly:

1. **Acceleration.** Is population vocabulary growth super-linear in cumulative input?
2. **Variability.** Do children differ in *how* and *when* their growth accelerates, not just in baseline ability?
3. **Input share.** How much of the between-child variance is attributable to input rate, vs. to internal factors like efficiency?

Does the same machinery applied to large language models yield the same answers? **Spoiler:** no — LLMs show a categorical disanalogy with children's learning. (Later in the talk.

Outline

1. Building the model: individual variability in acceleration
2. Input quantity: how much does it really vary?
3. Results: English and Norwegian longitudinal fits
4. Extensions: directly observed input and processing
5. Connecting to LLMs
6. Synthesis

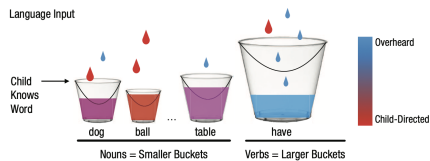
The accumulator hypothesis

The shared idea. Each child accumulates word-evidence at her own input rate r_i ; word j is acquired once its difficulty δ_j is crossed.

Ability scales as a power of input. If kids vary only in *rate*, then ability grows as $(\text{tokens})^1$ — the pure *unit accumulator*. Between-child differences come only from differences in r_i .

What our model will check.

- Is the growth exponent really 1, or steeper?
- Are children all alike except for input rate, or do they also differ in efficiency, in growth rate, and in when growth starts?



Bucket-fill metaphor: each child hears tokens at rate $r_i \cdot p_j$; word j is acquired when its bucket fills past difficulty δ_j (Kachergis et al. 2021).

Prior accumulator models: structural lineage

McMurray (2007). Pure accumulator. Each word w has acquisition threshold τ_w from population distribution $g(\tau)$. Acceleration requires $g'(\tau) > 0$.

Mitchell & McMurray (2009). McMurray + leverage feedback $a(t) = t + cL(t)$. *Shifts* growth without creating acceleration.

Hidaka (2013). Closed-form AoA distributions: cumulative (Gamma), rate-change (Weibull, exponent D on time), or both.

Mollica & Piantadosi (2017). Per-word Gamma waiting time: $\text{AoA} \sim \Gamma(k_w, \lambda_w)$.

Kachergis et al. (2021). 1-PL Rasch IRT with per-child θ_i and linear age covariate.

Each uses different notation. After we introduce our model, we'll re-express each in our coordinates and compare structurally.

The model: Rasch IRT + accumulator dynamics

$$P(\text{produces}_{ij}) = \sigma(\eta_{ij}) \quad \text{with} \quad \eta_{ij} = \theta_i - \delta_j$$

$$\eta_{ij} = \theta_i - \delta_j$$



kid ability
(grows with experience)

word difficulty
(per-word, free)

This is a standard Rasch IRT model. The substantive content is in how θ_i depends on accumulated experience. We build that up in three steps.

Step 1: pure accumulator

$$\theta_i = \log r_i + \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = r_i \cdot t$$

$$\theta_i = \log r_i + \log t$$



per-child input rate *time = age*
(words per month) *(months since onset)*

Ability = rate × time. The only thing that changes across development is more accumulated input. The only source of variation between children is variation in the input rate r_i .

Step 2: + per-child efficiency α_i

$$\theta_i = \log r_i + \log \alpha_i + \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = \alpha_i \cdot r_i \cdot t$$

$$\theta_i = \log r_i + \log \alpha_i + \log t$$



input rate *per-child efficiency (NEW)* *time*

Each child has a multiplicative efficiency α_i over and above their input rate. **The question we'll ask of the fit:** how much of the between-child variance comes from efficiency vs. input rate?

We summarize this share as $\pi_\alpha = \sigma_\alpha^2 / (\sigma_\alpha^2 + \sigma_r^2)$.

Step 3: + population acceleration κ_{pop}

$$\theta_i = \log r_i + \log \alpha_i + \kappa_{\text{pop}} \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = \alpha_i \cdot r_i \cdot t^{\kappa_{\text{pop}}}$$

$$\theta_i = \log r_i + \log \alpha_i + \kappa_{\text{pop}} \cdot \log t$$



input rate *efficiency* **population
scaling
exponent
(NEW; shared)** *time*

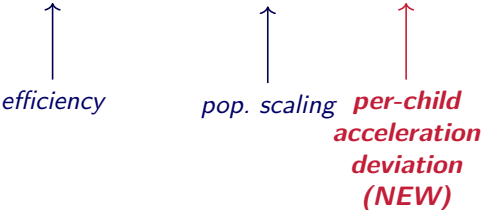
The population accelerates: ability scales as $t^{\kappa_{\text{pop}}}$ super-linearly. All children still share one κ_{pop} .

What we'll ask: is κ_{pop} really above 1 — i.e., does ability grow faster than a unit accumulator predicts? This is Hidaka's rate-change model.

Step 4: + per-child acceleration ζ_i

$$\theta_i = \log r_i + \log \alpha_i + (\kappa_{\text{pop}} + \zeta_i) \log t$$

$$\theta_i = \log r_i + \log \alpha_i + (\kappa_{\text{pop}} + \zeta_i) \cdot \log t$$



efficiency *pop. scaling* ***per-child
acceleration
deviation
(NEW)***

Each child has her own growth exponent $\kappa_i = \kappa_{\text{pop}} + \zeta_i$. The new piece: between-child variation isn't just in baseline ability, but in *how trajectories accelerate*. Every prior accumulator model forces every child to share one growth rate. **What we'll ask:** do children really differ in how fast their vocabulary accelerates?

Step 5: + per-child trajectory phase s_i

$$\theta_i = \log r_i + \log \alpha_i + (\kappa_{\text{pop}} + \zeta_i) \log(t - s - s_i)$$

$$\theta_i = \dots + (\kappa_{\text{pop}} + \zeta_i) \cdot \log \left(\overset{\text{calendar age}}{\underset{\uparrow}{t}} - s - \overset{\text{per-child trajectory phase (NEW)}}{\underset{\uparrow}{s_i}} \right)$$

calendar age *per-child trajectory phase (NEW)*

The same kind of individual-variability addition, but on the *time* dimension: children differ in *when* their developmental clock effectively starts. s_i can be positive (later start) or negative (developmentally ahead). **What we'll ask:** does the timing of growth onset really vary across children, alongside how fast they accelerate?

What every prior model pins; what we free

Model	efficiency σ_α	pop. growth κ_{pop}	per-child growth σ_κ	per-child phase σ_s	Per-word $\delta_j?$
McMurray (2007)	0	1	0	0	free, $\delta_j = \log \tau_j$
Mitchell & McMurray (2009)	0	1*	0	0	free, plus leverage c
Hidaka (2013) cumulative	0	1	0	0	free, $\delta_j = \log(N_j/\delta)$
Hidaka (2013) rate-change	0	$D+1$	0	0	free
Mollica & Piantadosi (2017)	0	1	0	0**	free, $\delta_j = \log(k_j/\lambda_j)$
Kachergis et al. (2021)	free	free	0	0	free
Best-fitting model (this work)	free	free	free	free	free

What's structurally new. Every prior model pins σ_κ and σ_s to zero (same growth rate and onset for all kids). Our model frees both; we find $\sigma_\kappa \approx 3.5$ and $P(\kappa_i > 1) > 0.997$ — the load-bearing additions that let the model fit acceleration and variability *jointly*.

** Mitchell & McMurray's leverage isn't linear in $\log t$. ** Mollica's Gamma encodes a per-word offset, but shared across kids. See appendix for each model translated individually.*

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Where does σ_r come from? The input-rate prior

The kid-level intercept $\xi_i = \log r_i + \log \alpha_i$ has total variance σ_ξ^2 that the data *does* identify. But the split between input rate (σ_r^2) and efficiency (σ_α^2) is not separately identified within the model:

$$\sigma_\xi^2 = \sigma_r^2 + \sigma_\alpha^2 \quad \Rightarrow \quad \pi_\alpha = 1 - \frac{\sigma_r^2}{\sigma_\xi^2}$$

We resolve this by pinning σ_r externally — from data quantifying actual variation in child-directed input rates across families. The model then partitions the data-identified σ_ξ^2 into input and efficiency components.

The pin matters for π_α .

- Bigger $\sigma_r \Rightarrow$ more variance assigned to input $\Rightarrow$ lower π_α .
- Pinning σ_r *too low* would overstate efficiency's share; *too high* would mask it.

Strategy. Compile the empirical literature on adult-token rates per child, derive a defensible σ_r , and report a full sensitivity sweep so the headline doesn't hinge on the central pin.

Empirical estimates of input-rate variation

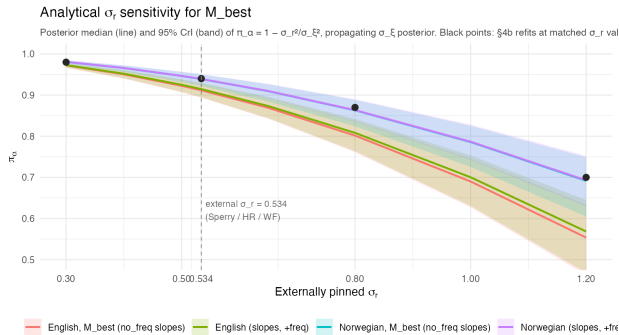
Group	n	Mean (tok/hr)	± 1 SD range (tok/hr)	Tokens/month mean (range)	σ_r (log)
Western, CDS-mother (Sperry sites, HR, WF)	17	$\sim 1,100$	620–2,000	400K (230K–730K)	0.57
Western, all-adult tokens	9	$\sim 2,200$	1,450–3,400	800K (530K–1.24M)	0.43
Sperry within-pool, CDS-any-adult	42	$\sim 1,500$	880–2,620	547K (320K–956K)	0.53
All groups (incl. Tseltal, Yéli, Pirahã)	27	—	~ 300 –5,000	(~ 110 K–1.8M)	1.24

Central pin: $\sigma_r = 0.534$ from the Sperry within-pool. A typical Western child hears ~ 547 K adult tokens/month (± 1 SD: ~ 320 K–960K, a $2.5\times$ spread). The Wordbank CDI:WS sample is overwhelmingly WEIRD, so the cross-cultural 1.24 is an outer bound, not a likely value.

Tokens/month = tokens/hr $\times$ H, H = 365 hr/mo (12 waking hr/day $\times$ 30.44 days/mo, model convention).

Sources: Sperry et al. 2019, Hart & Risley 1995, Weisleder & Fernald 2013, Bergelson 2018/2019/2023, Bunce et al. 2024, Casillas 2020/2021, Coffey et al. 2024.

Sensitivity: π_α across plausible σ_r



$\pi_\alpha(\sigma_r) = 1 - \sigma_r^2/\sigma_\xi^2$, with σ_ξ fixed at its data-identified posterior.

Central pin $\sigma_r = 0.534$: $\pi_\alpha \approx 0.91-0.94$.

Doubled $\sigma_r = 1.2$ (cross-cultural upper bound): π_α still > 0.55 .

Halved $\sigma_r = 0.30$ (below any defensible single-site SD): $\pi_\alpha \approx 0.97$.

Conclusion. The efficiency-dominated decomposition holds across the full range supported by Western input data, and the qualitative answer survives even the cross-cultural upper bound.

Caveat: this is about input *quantity*, not *quality*

What we've modeled. Variation in input *quantity* — how many adult tokens per hour each child hears. Under this lens, between-child differences in vocabulary growth are mostly *not* explained by input quantity.

What we haven't modeled. Variation in input *quality* — diversity, contingency, conversational structure, child-directed register. Quality variation is real and may matter for learning.

Why this is still a defensible scope:

- **Quality is hard to measure at scale.** No cross-population benchmark analogous to LENA-style token rates.
- **Quality would have to be enormous to escape the finding.** A per-child age-varying quality multiplier $q_i(t) = (t/a_0)^{\gamma_i}$ is observationally equivalent to a steeper κ_i — so to absorb the observed acceleration, “quality” would need to grow $\sim 3\{,\}000\times$ for the median kid and $\sim 10^6\times$ for the 95th percentile between 12 and 30 mo. (See appendix.)

Bottom line. Quantity is too small and too well-measured to explain the variability; quality would have to be implausibly large and age-varying to substitute.

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Data and compute for the longitudinal fits

Data.

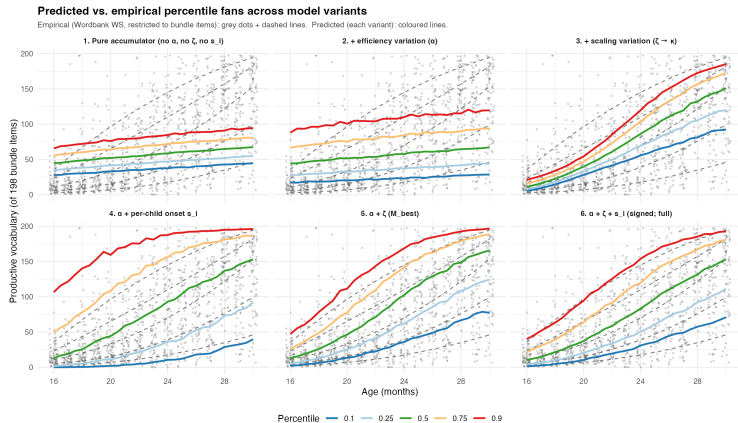
- **English Wordbank (WS form).** $I = 500$ children, ~ 1500 administrations, $J = 671$ items (full WS list), $N \approx 1.1\text{M}$ observed responses. Stratified-by-median-age sampling.
- **Norwegian Wordbank.** $I = 500$ children, ~ 3000 administrations, $J = 716$ items, $N \approx 2.2\text{M}$ responses. Same stratification.

Model family. The same five-step build run as a sequence of Stan models on each dataset: pure, $+\alpha$, $+\kappa_{\text{pop}}$, $+\zeta_i$, $+s_i$. Each fit warm-starts from the previous one's posterior.

Compute. Two GCP virtual machines (n2d-standard-128, 64 physical AMD EPYC cores each, 128 GB RAM), one per language, running in parallel. Stan with `reduce_sum` threading: 4 chains $\times$ 16 threads/chain saturates the 64 physical cores per machine. $\sim 11\text{h}$ per fit, $\sim 55\text{h}$ per VM for the five-variant family; the two VMs run in parallel, so $\sim 55\text{h}$ wall-clock and $\sim 110\text{h}$ of 128-core VM time total. No GPUs.

Priors and pins. $s = 6$ mo (population trajectory anchor); $\sigma_r = 0.534$ (Western-CDS empirical pin, see previous slides). Posteriors with $\hat{R} < 1.05$ on all parameters of interest.

Each addition matters: 5-panel comparison to data

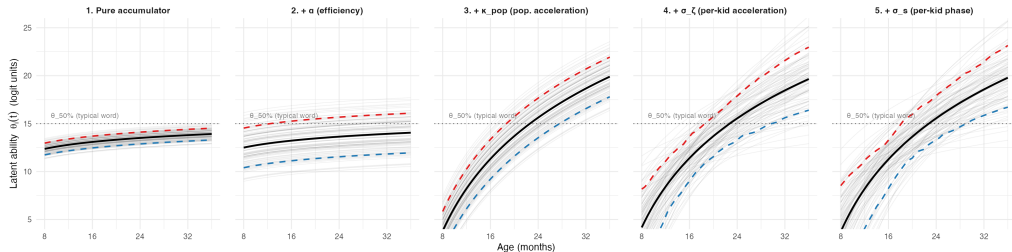


Reading. Each panel adds one mechanism. Pure accumulator collapses kids to one trajectory; σ_α adds vertical spread; κ_{pop} adds curvature; σ_ζ opens the fan with age; σ_s handles the floor at young ages.

Same story in θ -space

Predicted latent ability across the 5-model build

Each panel: 100 simulated kids (grey), population mean (black), 10th/90th percentiles (blue/red dashed). Dotted: θ where a typical-difficulty word has 50% acquisition probability. Posterior values from current English fits ($s=0.5$); will refresh after $s=6$ refit.



Same five-panel build, but plotting latent ability $\theta_i(t)$ directly instead of integrating to vocabulary. Population mean (bold), 10th/90th percentiles (blue/red), 100 simulated kids (faint). Dotted: $\theta_{50\%}$, the value at which a typical-difficulty word has 50% acquisition probability.

What it shows. The first two panels never cross $\theta_{50\%}$ within the data range — pure accumulators (with or without efficiency variation) are structurally unable to fit children's acquisition. Each subsequent panel adds a structural mechanism the data demands.

Best-fitting model: parameter posteriors

Parameter	Posterior	What it does
σ_α	1.71 [1.55, 1.91]	Between-child efficiency variation.
π_α	0.91 [0.89, 0.93]	Share of intercept variance from efficiency, not input.
κ_{pop}	10.35 [10.21, 10.51]	Population scaling exponent: ability scales as (tokens)^{κ_{pop}}.
σ_ζ	3.47 [3.14, 3.84]	Between-child variation in scaling exponent.
σ_s	[refits pending]	Between-child variation in trajectory phase.
$\rho(\xi, \zeta)$	+0.35 [+0.22, +0.47]	Higher-baseline kids have steeper growth.

Variability [0.2em] $\pi_\alpha = 0.91$:
efficiency variance dominates
input variance.

Acceleration [0.2em]
 $\kappa_{\text{pop}} \approx 10.4 \gg 1$: scaling is
wildly super-linear, far from
the unit accumulator.

Variation in scaling [0.2em]
 $\sigma_\kappa > \sigma_\alpha$: kids vary more in scaling
exponent than in baseline.

Cross-language acceleration

Sample	admins/kid	κ_{pop}	σ_{κ}	kid range ($\kappa_{\text{pop}} \pm 2\sigma_{\kappa}$)
English	median 3	10.4 [10.2, 10.5]	3.5 [3.1, 3.8]	~ 3 to 17
Norwegian	median 8	12.4 [12.3, 12.6]	3.7 [3.4, 4.2]	~ 5 to 20

$\kappa_{\text{pop}} \sim 10-12$ in both languages, with $P(\kappa_i > 1) > 0.997$.

δ vs. ζ_i decomposition is identifiability-dependent. In Norwegian (8 admins/kid), per-child slopes are well-identified — dropping ζ_i hurts fit by 235 ELPD. In English (3 admins/kid), δ pools what Norwegian's data lets fall into ζ_i .

Reading. The acceleration signature is universal. Its internal parameterization is identifiability-dependent on data density.

In any dense longitudinal sample, between-child κ_i spans roughly 5–15.

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Three model extensions: testing π_α with richer data

The Wordbank longitudinal fits use *population-level* input rate priors (Sperry/Hart-Risley/Weisleder-Fernald). Two natural follow-ups:

1. **Per-child observed input:** use *measured* per-child input rate instead of a population prior. Does π_α stay ~ 0.9 when input is no longer a pooled estimate? → **BabyView**, **SEEDLingS**.
2. **Per-child processing efficiency:** link CDI growth to an independent per-child readout of processing speed. Does the σ_α we estimate from CDI reflect “processing efficiency” or something else? → **Peekbank**.

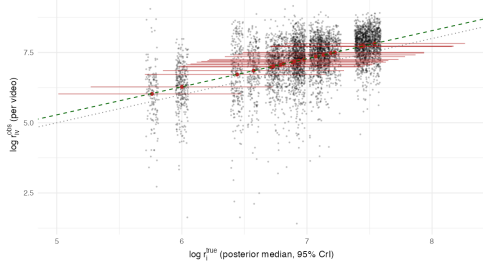
Both extensions reuse the same model machinery with an added channel in the linear predictor; only the data and the per-child latents change.

Replicating with directly observed input

BabyView, $N = 20$

Per-child observed input vs inferred true rate

Dashed green: $y = x + \text{beta_react}$ ($= 0.28$); dotted: $y = x$, Reactivity inflation = 33% (CrI: -35-173%)

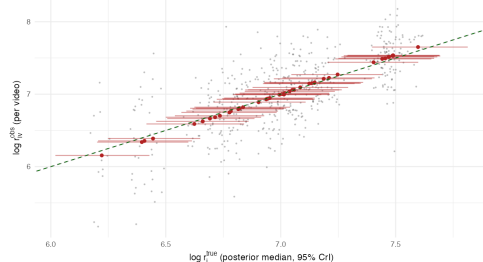


$$\pi_{\alpha} = 0.84, [0.68, 0.93]$$

SEEDLingS, $N = 44$ (comp corrected)

Per-child observed input vs inferred true rate

Dashed green: $y = x + \text{beta_react}$ ($= -0.00$); dotted: $y = x$, Reactivity inflation = -0% (CrI: -0-0%)



$$\pi_{\alpha} = 0.94, [0.89, 0.97]$$

With per-recording observed input, the π_{α} headline holds: input-rate variance is a small share of between-child variance even when measured directly.

Adding a processing-speed readout: Peekbank LWL

Joint model adds a per-child LWL log-RT readout coupled to $\log \alpha_i$ via γ_{rt} .

Posterior.

- $\gamma_{rt} = 0.082$, [0.046, 0.125] — LWL processing speed is real, and bounded above zero.
- $\pi_{\alpha} = 0.90$, [0.86, 0.94] — replicates within the Peekbank subset.
- $\rho(\kappa_i, \text{rtslope}_i) = -0.02$, [-0.33, +0.27] — CDI-side and LWL-side growth rates do *not* share variance.

Two clocks, not one. The acceleration in vocab growth and the maturation of processing speed are independent maturation processes within the same child — different “efficiency clocks” ticking in parallel.

This rules out the simplest unification (“one underlying efficiency axis”); the two readouts each pick up a different developmental process.

π_α across five samples

Sample	N	σ_α	π_α	Channel
English longitudinal	200	1.83 [1.65, 2.04]	0.91 [0.90, 0.93]	Pooled input prior
Norwegian longitudinal	200	2.10 [1.90, 2.35]	0.94 [0.93, 0.95]	Pooled input prior
Peekbank (Stanford)	62	1.64 [1.34, 2.03]	0.90 [0.86, 0.94]	Pooled + LWL
BabyView	20	1.13 [0.86, 1.61]	0.84 [0.68, 0.93]	Head-mounted video
SEEDLingS	44	1.39 [1.13, 1.74]	0.94 [0.89, 0.97]	LENA all-day audio

$\pi_\alpha \in [0.84, 0.94]$ **across all five samples.** The specific input channel (pooled vs. measured) and sample size hardly move the picture: between-child variance is dominated by something *other* than input rate.

(Placeholder: numbers from $s=0.5$ fits; $s=6$ refits running. The qualitative pattern is invariant.)

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Why look at LLMs?

LLMs and children both learn word meanings from input. They differ *enormously* in scale and architecture, but the question of how their learning scales with input is statistically well-posed:

- For an LM and a word w , we have $P(w \text{ produced correctly})$ as a function of training tokens.
- This is a sigmoid in log-tokens with a per-word slope.
- The same statistic exists for human children: $P(w \text{ acquired})$ as a function of cumulative input.

This lets us put kids and LMs on the same axis — not just trade anecdotes about scaling. Two questions:

1. Do kids and LMs have similar per-word acquisition slopes?
2. Does a model that fits kids' learning say anything about LMs?

Chang & Bergen (2022): per-word sigmoids in LMs

Setup. Chang & Bergen trained four LMs (BERT, biLSTM, GPT-2, LSTM) on the same input corpus (BookCorpus + WikiText). At each of ~ 200 training checkpoints, they probed each of ~ 611 CDI words to get $P(\text{word produced correctly} \mid \text{checkpoint})$.

For each (LM, word) they fit a 4-parameter logistic

$$S_w(x) = \text{lower}_w + \frac{\text{upper}_w - \text{lower}_w}{1 + \exp((x - \text{mid}_w)/\text{scale}_w)}$$

where $x = \log_{10}(\text{training steps})$, yielding a per-word **scaling slope**: how fast that word's acquisition probability rises with cumulative training. After unit conversion, this is exactly analogous to our $\kappa_i \log t$ for children.

Result they reported. Across the four LMs, mean per-word slope ≈ 1 logit per nat-log training-step. Striking but ambiguous: is this the LM architecture or the input distribution?

Could the gap be an input-distribution artifact?

Confound in C&B's setup. Their LMs trained on BookCorpus + WikiText (adult written text), not on the speech kids actually hear. So the slope gap conflates two things: (a) a structural difference in how LMs and kids scale, and (b) an input-distribution mismatch.

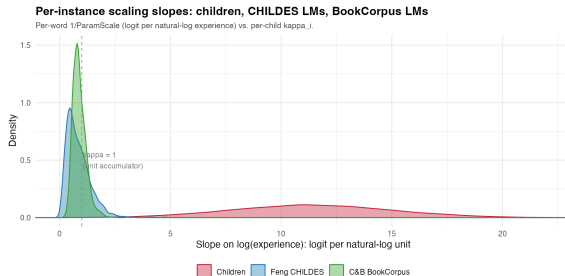
Test: retrain GPT-2 small on CHILDES (the corpus kids actually hear, per Feng et al. 2026), keep everything else fixed, refit C&B's per-word sigmoid. If the gap is input-driven, CHILDES-trained slopes should rise toward children.

Setup:

- GPT-2 small (124M params, $12L \times 768d \times 12h$)
- CHILDES train split, $\sim 24.5M$ tokens, 20 epochs (Feng et al. §B spec)
- 3 random seeds ($\{0, 42, 123\}$) as a between-instance variance probe
- 73 log-spaced eval checkpoints; per-word mean NLL over 50 occurrences in the CHILDES validation set
- C&B 4-parameter logistic fit per (LM, word); per-word slope = $0.434 / \text{ParamScale}_w$
- 609 / 611 C&B CDI words are single tokens in the GPT2-CHILDES BPE

Single L40S 48GB per seed (Stanford Sherlock gpu partition), $\sim 8h$ wall-clock each.

The per-word view: CDS training does not move the LM slope

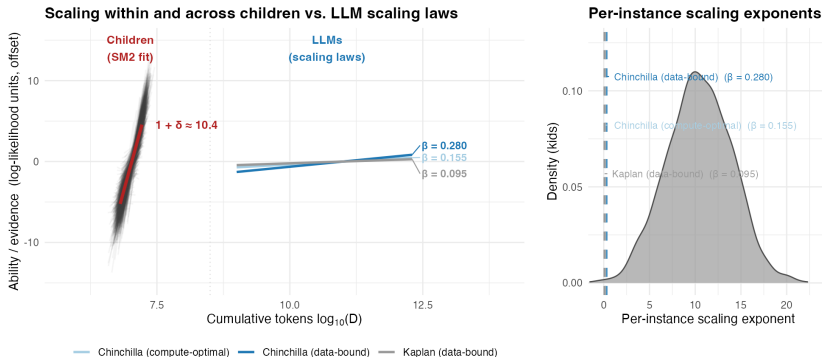


Population	N	Median	IQR
Children (English best-fit)	5000	10.3	[8.0, 12.6]
GPT-2-CHILDES (3 seeds)	609 ea	0.72–0.74	[0.43, 1.16]
GPT-2-BookCorpus (C&B)	604	0.81	[0.45, 1.54]
BERT, biLSTM, LSTM-BookCorpus	593–609	0.76–0.96	—

CHILDES-trained GPT-2 (0.72–0.74) sits inside the BookCorpus cluster (0.76–0.96), all $\sim 14\times$ below kids (10.3). *The input-distribution explanation accounts for ~ 0 of the kid-vs-LM gap; the gap is structural.*

The aggregate view: scaling laws

Slopes are the structurally meaningful comparison; y-axis offset is anchored for visual clarity. Kid D range comes from per-kid r_i .



SM2 fit long_slopes: median $1+\delta = 10.39$ [10.24, 10.55]; $\sigma_\zeta = 3.48$ [3.15, 3.87]; $\mu_{r_i} = 7.34$, $\sigma_{r_i} = 0.53$, $a_i = 19$. Chinchilla and Kaplan exponents from Hoffmann et al. 2022 / Kaplan et al. 2020.

Comparing two estimates of *ability gained per token of experience*: LM scaling laws (Kaplan, Chinchilla compute-optimal, Chinchilla data-bound) all have exponents $\beta \in [0.10, 0.28]$. Children's $\kappa_{\text{pop}} \approx 10$ is two orders of magnitude steeper — a different scaling regime, not a quantitative degree.

Outline

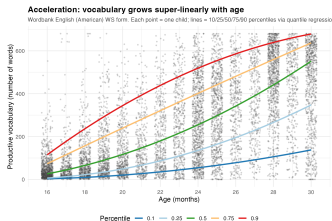
1. Building the model: individual variability in acceleration
2. Input quantity: how much does it really vary?
3. Results: English and Norwegian longitudinal fits
4. Extensions: directly observed input and processing
5. Connecting to LLMs
6. Synthesis

Technical summary

- **Acceleration is real and population-level.** $\{(placeholder\ for\ s=6\ refits)\}$ $\kappa_{pop} \approx 10$, confidently above the unit-accumulator boundary $\kappa_{pop} = 1$. The first prior model to free κ_{pop} (Hidaka rate-change) was structurally right.
- **Individual variability in acceleration is real too.** $\sigma_{\zeta} > 0$ and $\sigma_s > 0$: children differ not just in baseline ability, but in *how their trajectories accelerate* and *when*. No prior accumulator model frees these.
- **Input quantity is a small share of between-child variance.** $\pi_{\alpha} \approx 0.91$ across 5 samples. Aligns with the meta-analytic $R^2 = 0.04\text{--}0.07$ of Coffey & Snedeker (2026).
- **LLMs are structurally different.** Same per-word sigmoid statistic gives $\kappa \sim 1$ for LMs vs. ~ 10 for children — a $10\times$ gap with near-zero between-instance variance in LMs.

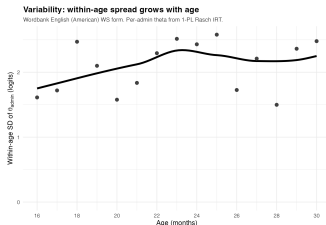
github.com/langcog/standard-model-2

Two signatures, twin constraints on theory



Acceleration. $\kappa_{\text{pop}} \approx 10$ — above any pure accumulator. *Where does the super-linearity come from?*

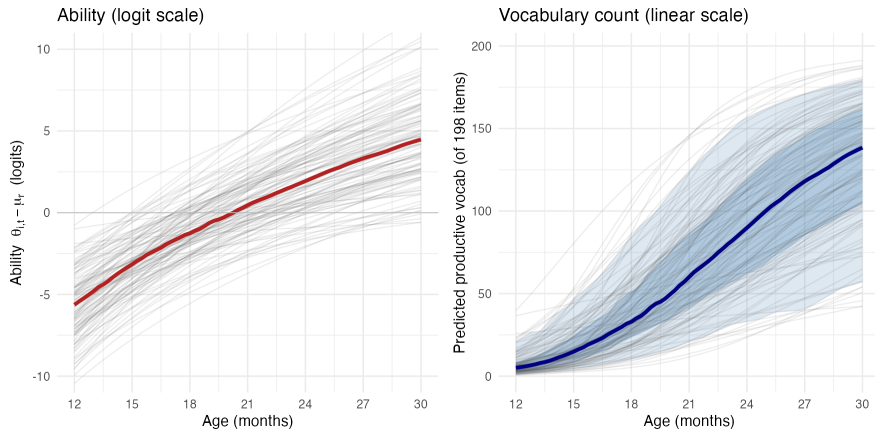
Twin constraints on any theory: **(a)** population-level super-linear growth (maturation? learning-to-learn?); **(b)** individual differences *not* in input quantity (processing? interactional quality?). *Pure accumulators satisfy neither.*



Variability. $\pi_{\alpha} \approx 0.91$: input explains $\sim 10\%$ of between-child variance. *Where does the other 90% live?*

Appendix

Appendix: per-child trajectories from the best fit



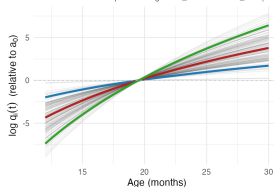
120 kids sampled from the best-fitting model's posterior. **Left:** latent ability θ (logit). **Right:** predicted vocabulary. Concave-up shape = super-linear signature; fan width = between-child variability.

Appendix: σ_r sensitivity from earlier cross-sectional fits

Quality-multiplier interpretation: same model, different label

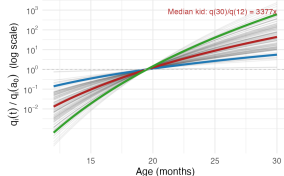
Implied per-child quality multipliers $q_i(t)$

Under the structural equivalence: $\gamma_i = \delta + \zeta_i \sim N(9.35, 3.47^2)$. Quality is a multiplier on the token quality, this is how much quality is needed to absorb the acceleration.



kid percentile — 5% — 50% — 95%

Same trajectories on multiplicative scale



kid percentile — 5% — 50% — 95%

$\delta + \zeta_i \sim N(9.35, 3.47^2)$. For 'token quality grows with age' to absorb M_best's acceleration, median kid needs $q(30)/q(12) \sim 3377\times$; ± 1 SD spread covers $40\times$ to $981046\times$.

Reviewer reflex. "Maybe kids' tokens differ in *quality*, and that's what looks like acceleration."

Structural answer. Per-child age-varying quality $q_i(t) = (t/a_0)^{\gamma_i}$ on input is observationally equivalent to $\gamma_i = \kappa_i - 1$.

Implication. Quality cannot escape acceleration — it *relabels* it. For the median kid, "quality" would have to grow $\sim 3,000\times$ from age 12 to 30 mo. For the 95th percentile kid, $\sim 10^6\times$. Implausible as a substantive quality story.

Appendix: McMurray (2007) in our notation

Original. Each word w has threshold $\tau_w \sim g(\tau)$; deterministic step at $t \geq \tau_w$. Vocabulary $L(t) = G(t)$ (CDF of g). Acceleration requires $g'(\tau) > 0$ over the productive window.

In our notation:

$$\eta_{ij}(t) = \log t - \log \tau_j = \kappa \log t - \delta_j \quad \text{with} \quad \kappa \equiv 1, \quad \delta_j = \log \tau_j.$$

Parameter	Setting in McMurray
$\log r_i$	$\equiv 0$ (no per-child rate variation)
$\log \alpha_i$	$\equiv 0$ (no efficiency variation)
κ_i	$\equiv \mathbf{1}$ (unit accumulator)
δ_j	free, drawn from g

Appendix: Hidaka (2013) in our notation

Original. Three families: cumulative (N -accumulator at constant rate, $\text{AoA} \sim \Gamma$), rate-change (1-accumulator at rate $\delta \cdot t^D$, $\text{AoA} \sim \text{Weibull}$), and both. **Rate-change:** ability $\propto t^{D+1}$; $D > 0$ produces super-linear scaling.

In our notation (rate-change form):

$$\eta_{ij}(t) = (D + 1) \log t - \delta_j \quad \text{with} \quad \kappa_i \equiv D + 1.$$

Parameter	Setting in Hidaka rate-change
$\log r_i$	$\equiv \log \delta_{\text{const}}$ (shared population rate)
$\log \alpha_i$	$\equiv 0$ (no efficiency variation)
κ_i	$\equiv \mathbf{D + 1}$ (shared, no per-child variation)
δ_j	free

Hidaka's $D = \kappa_{\text{pop}} - 1$ in our notation. His $D > 0$ is our $\kappa_{\text{pop}} > 1$.

Appendix: Mollica & Piantadosi (2017) in our notation

Original. Per-word: k_w effective learning instances arrive as Poisson at rate λ_w ; word acquired after k_w events. $AoA_w \sim \Gamma(k_w, \lambda_w)$. For large k_w , Gamma CDF approximates Gaussian (mean k_w/λ_w) and then a ~ 1.7 -scale logistic.

In our notation:

$$\eta_{ij}(t) = \log t - \delta_j \quad \text{with} \quad \kappa_i \equiv 1, \quad \delta_j \approx \log(k_j/\lambda_j).$$

Parameter	Setting in Mollica & Piantadosi
$\log r_i$	$\equiv \log \lambda$ (shared population rate)
$\log \alpha_i$	$\equiv 0$ (no per-child variation)
κ_i	$\equiv \mathbf{1}$ (unit-rate accumulator)
δ_j	$\approx \log(k_j/\lambda_j)$, free per-word

Appendix: Kachergis et al. (2021) in our notation

Original. 1-PL Rasch IRT with per-child latent ability:

$\text{logit } P(y_{ij} = 1) = \theta_i + d_j + \beta_a a_i + \beta_t \log(p_j H a_i)$. Per-child θ_i free; d_j per-word free; shared age slope β_a across all kids.

In our notation:

$$\eta_{ij} = (\log r_i + \log \alpha_i) + \kappa \log a_i - \delta_j \quad \text{with} \quad \kappa \equiv \text{shared population slope.}$$

Parameter	Setting in Kachergis 2021
$\log r_i$	free, no per-child variance specified
$\log \alpha_i$	free (their θ_i absorbs $\log r + \log \alpha$)
κ_i	$\equiv$ shared population slope (no per-child variance)
δ_j	free (their d_j)

First prior model to free σ_α . But σ_κ is still zero — every kid has the same scaling slope.